

TRIssecure: Design and Evaluation of a Three-Factor Post-Quantum Biometric E-Voting Framework on Commodity Edge Hardware

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Abstract

Electronic voting systems must simultaneously satisfy strong voter authentication grounded in robust signal-level biometric evidence, tamper-evident vote integrity, end-to-end voter verifiability, and practical deployment constraints on resource-limited hardware. Existing solutions treat these requirements in isolation: biometric systems omit formal security verification; blockchain-based systems ignore biometric authentication; and no existing system addresses emerging quantum threats while remaining deployable on edge hardware. TRIssecure V2 is a three-factor biometric e-voting terminal that jointly satisfies all four requirements on a commodity Raspberry Pi 4 (ARM64, commodity edge hardware) without requiring server-grade computing or persistent cloud connectivity. It integrates an ISO/IEC 19795-1 compliant biometric pipeline with Bayesian signal-level fusion of face-recognition similarity scores and NFC channel evidence, ISO/IEC 30107-3 presentation attack detection via spoof-signal classification (MiniFASNetV2 liveness), a hybrid post-quantum cryptographic layer (Kyber-768+X25519 KEM and Dilithium-3+Ed25519 signatures per NIST FIPS 203/204), and a formally verified five-stage DFA authentication protocol with six proved security properties. A novel Adaptive Trust Score (ATS) engine dynamically adjusts component weights based on contextual risk signals (attempt history, temporal anomalies, location flags), escalating borderline cases for supervisor review rather than binary accept/reject—achieving 99.3% HIGH

trust on genuine users and zero Low classifications for simulated impostors under Gaussian score assumptions. Simulation-based evaluation (ISO/IEC 19795-6, $n=1\,000$ score pairs) achieves $EER = 1.55\%$, $AUC = 0.9991$, $TAR@1\%FAR = 97.7\%$, and $PAD\ ACER = 4.25\%$ (PAD layer only). A proof-of-concept pilot with $N=20$ participants on real hardware confirms $FAR=FRR=ACER=0.00\%$ across 271 authentication events (221 genuine, 35 impostor, 15 replay). Adversarial simulation across six attack categories confirms the defence stack reduces combined attack success from $\approx 35\%$ (face threshold alone) to zero simulated bypass events after the NFC-AES-GCM layer—a robust defence-in-depth property under standard cryptographic assumptions within the evaluated threat model. On-device benchmarking delivers 110.7 ms end-to-end authentication latency (p99 = 111.3 ms) and 541.6 voters/min throughput. To the best of our knowledge, no prior e-voting system simultaneously provides formal protocol verification, post-quantum cryptography, adaptive context-aware trust scoring, liveness detection, and end-to-end verifiability on low-cost edge hardware.

Keywords: biometric authentication, e-voting, formal verification, post-quantum cryptography, edge computing, presentation attack detection, Bayesian fusion, Merkle proof

1 Introduction

Electronic voting systems are critical infrastructure where failure consequences are asymmetric: a single undetected fraud vector can undermine election legitimacy at scale. The fundamental security challenge is to authenticate the right voter, record the right vote, prevent tampering, and allow the voter to verify their participation—all while running on deployable, affordable hardware. At its core, voter authentication is a signal-processing problem: a face-recognition similarity score and a liveness (presentation-attack) classification are noisy, adversarially manipulable measurements that must be fused, thresholded, and combined with an independent physical-credential channel (NFC) to reach a reliable security decision under a bounded latency budget. TRIsecure V2 treats this signal-level evidence fusion as the central design problem, layering formal protocol verification, post-quantum cryptography, and end-to-end verifiability on top of it.

Existing systems leave at least one requirement unsatisfied. Fingerprint- and face-based voting systems [1–3] report biometric error rates without formal protocol verification or post-quantum (PQC) readiness. Blockchain-based approaches [4–6] provide strong vote integrity but lack biometric authentication and rely on classical cryptography vulnerable to quantum adversaries [7]; a comprehensive review of 81 blockchain voting systems by Huang et al. [8] confirms this gap. Industry multi-factor authentication standards (FIDO2/WebAuthn [9]) are not designed for the one-time-per-election voting constraint and do not incorporate formal liveness detection or voter-verifiable receipts.

To the best of our knowledge, no existing e-voting system jointly provides: (i) multi-factor biometric authentication with liveness detection, (ii) formally verified protocol security, (iii) post-quantum cryptographic readiness, and (iv) end-to-end voter verifiability—all on low-cost edge hardware without requiring cloud connectivity.

Contributions

We address these gaps with **TRIsure V2**:

C1. Formally verified authentication. A five-stage pipeline modelled as a DFA with six proved security properties: determinism, completeness, safety, liveness, double-vote prevention, and rate-limit enforcement.

C2. Bayesian biometric fusion. Log-likelihood-ratio fusion of face cosine similarity and NFC channel evidence (EER= 1.55%, AUC= 0.9991, TAR@1%FAR= 97.7%).

C3. Hybrid post-quantum layer. Kyber-768+X25519 KEM (FIPS 203) and Dilithium-3+Ed25519 signatures (FIPS 204); all six PQC operations measured

37 on-device via liboqs 0.15.0 (ARM64): ML-KEM-768 encaps 0.088 ms, ML-DSA-
38 65 sign 1.536 ms, auth-path total 2.289 ms—within the < 200 ms latency
39 budget.

40 **C4. ISO/IEC 30107-3 PAD.** MiniFASNetV2 liveness detection (BPCER= 1.0%,
41 ACER= 4.25%).

42 **C5. End-to-end verifiability.** SHA-256 receipts and $O(\log n)$ Merkle inclusion
43 proofs.

44 **C6. Edge evaluation.** Raspberry Pi 4 (ARM64, 4 GB): 110.7 ms latency (p99=
45 111.3 ms), 541.6 voters/min throughput, CPU 52.5 °C under load.

46 **C7. Adaptive Trust Score (ATS) engine.** A context-aware weighted fusion
47 model that adjusts face/NFC/PAD component weights dynamically based on
48 risk signals (attempt history, temporal anomalies, location flags), producing
49 HIGH/MEDIUM/LOW trust decisions. Genuine users achieve 99.3% HIGH
50 trust; 100% of simulated impostor attempts are rejected. To the best of our
51 knowledge, no prior e-voting system incorporates dynamic risk-adaptive trust
52 scoring.

53 **C8. Adversarial attack evaluation.** Simulation of six attack categories (print
54 photo, video replay, deepfake, NFC UID clone, session replay, database tamper)
55 quantifies per-layer defence contribution. Full defence stack reduces combined
56 success from $\approx 35\%$ to zero simulated bypass events after NFC-AES-GCM.

57 **C9. Proof-of-concept hardware pilot.** End-to-end validation on a Raspberry Pi 4
58 with $N=20$ participants across 271 authentication events (221 genuine, 35 impos-
59 tor, 15 replay): FAR=FRR=ACER= 0.00%, confirming system correctness on
60 real hardware without simulation.

61 No individual component is algorithmically novel in isolation. The **novel con-**
62 **tributions** are: (i) the ATS engine, which introduces dynamic risk-adaptive weight

adjustment absent from all prior biometric e-voting systems; (ii) their *principled co-design*; and (iii) the *complete systems security architecture*—a fully deployable secure voting terminal on low-cost edge hardware, validated end-to-end on real hardware with $N=20$ participants. To the best of our knowledge, no prior e-voting system delivers all six security properties simultaneously on a Raspberry Pi class device. In detail, the co-design provides: (i) Bayesian LR fusion of face and NFC modalities reduces the effective FAR by three orders of magnitude without increasing biometric EER—an attack requiring a valid NFC card *and* a face spoof simultaneously; (ii) the DFA verification executes *in the deployed code* (not as an offline proof), so removing it breaks pipeline functionality, creating a tight coupling between formal security and operational correctness; and (iii) all five mechanisms—fusion, PAD, PQC, DFA verification, and Merkle verifiability—fit within the latency (< 200 ms) and resource budget of commodity edge hardware, demonstrated on a Raspberry Pi 4.

2 Related Work

2.1 Biometric E-Voting Systems

Bello et al. [1] achieve $EER = 2.5\%$ at 950 ms authentication latency with no PAD, PQC, or formal verification. Komatineni and Lingala [2] combine machine-learning face detection with biometric scanning for duplicate-vote prevention in a two-factor architecture. Nagaraj et al. [3] integrate facial recognition with fingerprint authentication to strengthen polling station security. Megalingam et al. [10] combine a voter identification card with fingerprint biometric authentication, achieving low-cost deployment on constrained hardware but providing no liveness detection, formal protocol verification, or post-quantum cryptography. Varma and Megalingam [11] demonstrate FPGA-accelerated biometric voting, showing that dedicated silicon inference reduces latency below ARM64 software execution, but the system lacks post-quantum cryptographic support and end-to-end verifiable receipts. Revathy et al. [12]

89 integrate CNN-based face recognition with blockchain and blind signatures, showing
 90 that combining deep learning classifiers with immutable vote records improves both
 91 authentication accuracy and tamper evidence. Ashok Kumar et al. [13] demonstrate
 92 an online smart voting platform using face recognition for voter identification. Faruk et
 93 al. [6] deploy blockchain and biometric verification (fingerprint+face) over Hyperledger
 94 Fabric, demonstrating 87% biometric registration success across 100 participants but
 95 relying on classical cryptography with no formal protocol verification. NIST SP 800-
 96 63B [14] defines three authenticator assurance levels (AAL1–3) but does not prescribe
 97 biometric liveness detection or formal protocol verification for voting contexts. Face
 98 recognition systems based on FaceNet [15], CosFace [16], and ArcFace [17] achieve
 99 sub-1% EER on LFW but require GPU-class hardware and include no voting infras-
 100 tructure. Wang and Deng [18] survey deep face recognition across 50+ architectures;
 101 none are integrated with formal verification or e-voting pipelines. George et al. [19]
 102 demonstrate EdgeFace, a 2024 compact model achieving sub-1% EER on a mobile
 103 ARM processor—confirming the feasibility trajectory of high-accuracy face recogni-
 104 tion on the same hardware class as TRIsure V2. Buchmann et al. [20] provide a
 105 2024 systematic review of remote e-voting security, identifying formal verification and
 106 biometric binding as the two most significant unresolved gaps across all deployed sys-
 107 tems. TRIsure V2 achieves competitive EER (1.55%) on ARM64 while additionally
 108 providing formal protocol guarantees and PQC readiness.

109 **2.2 Blockchain-Based Voting**

110 Fujioka et al. [21] proposed the foundational blind-signature-based secret ballot
 111 scheme; Benaloh and Tuinstra [22] formalised receipt-freeness. Nakamoto’s Bitcoin [23]
 112 demonstrated append-only hash chaining, inspiring blockchain voting proposals.
 113 Ayed [4] proposes Ethereum-based voting (> 2100 ms) with PIN-only authentica-
 114 tion. Hardwick et al. [5] and Hjálmarsson et al. [24] extend this with anonymisation

115 and Hyperledger Fabric, but provide no biometric gate or anti-spoofing. Durga et
 116 al. [25] propose a private and secure blockchain mechanism for online voting using
 117 smart contracts to enforce vote confidentiality and election integrity, demonstrating
 118 practical deployment but relying on classical cryptography with no biometric identity
 119 binding or formal protocol verification. Juels et al. [26] formalise coercion-resistance,
 120 a property TRIssecure V2 partially addresses through in-person biometric binding
 121 (Section 7.4). Huang et al. [8] survey blockchain applications in voting across 81 prior
 122 works (ACM Computing Surveys 2022), identifying voter authentication and anti-
 123 spoofing as the primary unresolved gaps. Hajian et al. [27] survey blockchain e-voting
 124 architectures with emphasis on smart-contract ballot processing (Electronics 2024),
 125 confirming that biometric identity binding is absent from current blockchain deploy-
 126 ments. TRIssecure V2 achieves equivalent vote integrity via hash chaining and Merkle
 127 proofs without blockchain overhead, while adding biometric authentication.

128 **2.3 Post-Quantum Cryptography on Edge**

129 Sikeridis et al. [28] show Kyber-768 and Dilithium-3 are feasible in TLS 1.3 on ARM
 130 devices via liboqs. Paquin et al. [29] benchmark liboqs across x86 and ARM, providing
 131 the reference figures against which our Pi 4 measurements are calibrated. Kannwischer
 132 et al. [30] characterise PQC on Cortex-M4, establishing Kyber-768 key generation
 133 under 100k cycles. Delvaux et al. [31] provide 2024 benchmarks of NIST finalists on
 134 Cortex-A class processors, confirming ML-KEM-768 key encapsulation under 0.5 ms
 135 on Cortex-A53—motivating our projected 0.1 ms estimate for the Cortex-A72 (higher
 136 IPC). NIST finalised ML-KEM (FIPS 203 [32]) and ML-DSA (FIPS 204 [33]) in
 137 2024. Shor’s algorithm [34] threatens RSA and ECC; Grover’s algorithm [35] halves
 138 symmetric key security—motivating hybrid PQC even for edge deployments. To the
 139 best of our knowledge, no prior e-voting system integrates a hybrid PQC layer on
 140 low-cost ARM64 edge hardware.

141 2.4 Presentation Attack Detection

142 ISO/IEC 30107-3 [36] defines the PAD evaluation framework (APCER, BPCER,
143 ACER). Zhang et al. [37] introduce MiniFASNetV2, a lightweight CNN suitable for
144 embedded deployment. Yang et al. [38] provide a 2023 survey of presentation attack
145 methods including deepfakes; deepfake attack penetration against MiniFASNet-class
146 detectors is estimated at 22% (post-PAD), requiring the additional NFC gate for com-
147 plete mitigation. Liveness detection has not previously been integrated into a formally
148 verified e-voting pipeline. Liu et al. [39] leverage depth maps; Jourabloo et al. [40] use
149 spoof-noise disentanglement. Zhang et al. [41] show that GAN-generated fake images
150 carry spectral artefacts introduced by the up-sampling pipeline, motivating frequency-
151 domain analysis as a complement to depth-based liveness. Key benchmark datasets
152 include NUAA [42] and OULU-NPU [43].

153 2.5 Formal Verification of Authentication Protocols

154 The Dolev–Yao model [44] formalises the adversary capable of intercepting and
155 replaying all network messages, providing the theoretical basis for protocol secu-
156 rity analysis. ProVerif [45] and Tamarin [46] target cryptographic secrecy and
157 authentication goals via the applied-pi calculus. Lowe [47] used model checking
158 (FDR) to find and fix an attack on the Needham–Schroeder protocol. DFA the-
159 ory, formalised in Sipser [48], provides decidable analysis for pipeline control-flow
160 properties. TRIssecure V2 uses automata-theoretic verification because the properties
161 of interest—ordering, completeness, determinism, termination, double-vote preven-
162 tion, rate-limit enforcement—concern pipeline control-flow rather than cryptographic
163 secrecy, the DFA executes *in the deployed code*, and all six properties are decidable
164 by BFS/DFS without an external solver, yielding human-auditable proofs embedded
165 in the production codebase.

2.6 Verifiable E-Voting and Biometric Template Protection

Helios [49] pioneered web-based open-audit voting via homomorphic encryption. Civitas [50] provides strong coercion-resistance using threshold cryptography. Scantegrity II [51] achieves optical-scan E2E verifiability through confirmation codes. Selene [52] introduces tracker-number-based verifiability with coercion mitigation. The Norwegian scheme [53] deployed return-code verifiability at national scale. All five provide strong verifiability properties but leave identity binding to external authentication infrastructure with no biometric gate. Küsters et al. [54] and Benaloh et al. [55] formalise individual and universal verifiability, which TRIssecure V2 satisfies via Merkle receipts. Dodis et al. [56] formalise fuzzy extractors for biometric key derivation; Jain et al. [57] survey cancelable biometrics per ISO/IEC 24745 [58]; Nandakumar and Jain [59] survey the gap between biometric template protection theory and practice. TRIssecure V2 protects face embeddings via AES-256-GCM with per-voter keys; extending to cancelable representations is identified as future work.

2.7 Multimodal Biometric Fusion

Ross and Jain [60] establish that score-level fusion outperforms unimodal baselines. Nandakumar et al. [61] derive optimal LR fusion under Gaussian density estimation, which TRIssecure V2 adopts. Fierrez-Aguilar et al. [62] extend this with quality-based weighting; Jain et al. [63] provide a comprehensive multimodal survey.

2.8 Comparison of Prior Work and Capability Gap Matrix

Table 1 summarises six reference systems against TRIssecure V2. TRIssecure V2 is the only system to satisfy all capability dimensions on edge hardware. Figure 1 presents the full gap matrix across eight systems and seven dimensions.

Table 1 Comparison of related e-voting and biometric authentication systems.
PAD = presentation attack detection; PQC = post-quantum cryptography; FV = formal verification;
E2EV = end-to-end verifiability.

Author (Year)	Method	Advantages	Limitations
Bello et al. (2021) [1]	Fingerprint + smart card	Two-factor; biometric binding; EER 2.5%	No PAD; no PQC; no FV; 950 ms; server-dependent
Ayed (2017) [4]	Ethereum blockchain, PIN-only	Vote receipts; tamper-evident ledger	No biometric gate; > 2100 ms; classical crypto
Hardwick et al. (2018) [5]	Blockchain with anonymisation	Voter anonymity; blockchain integrity	No biometric; no PAD; classical crypto
Deng et al. (2019) [17]	ArcFace + PIN	EER 0.76%; 45 ms (GPU)	No liveness; no FV; requires GPU server
FIDO2/WebAuthn (2019) [9]	Device-bound MFA	Industry standard; 190 ms	No PAD; no voter receipts; not per-election
TRIssecure (Ours)	V2 NFC + face + session, PAD, PQC, DFA	FV; EER 1.55%; PAD; PQC; E2EV; edge ARM64	Simulation-based biometrics; PoC pilot ($N=20$, FAR=FRR=ACER=0.00%)

3 System Architecture

3.1 Design Goals and Threat Model

TRIssecure V2 satisfies five goals: **G1** Authentication (three-factor voter binding); **G2** Vote integrity (tamper-evident, auditable); **G3** Voter privacy (embeddings encrypted at rest); **G4** Verifiability (individual and universal); **G5** Edge viability (< 200 ms, commodity ARM64).

The threat model covers: *remote attacker* (brute-force, replay, MitM); *physical attacker* (spoofed face, cloned NFC card, SPI bus eavesdropping, fault injection via power glitching); *insider attacker* (DB read/write but no biometric secret); *quantum adversary* (Shor’s algorithm against RSA/ECC); *supply-chain attacker* (malicious SD card image or Pi firmware).

Capability Gap Matrix: Prior Work vs. TRIssecure V2							
	Biometric Auth	Liveness Detection (PAD)	Formal Verification	Post-Quantum Crypto	End-to-End Verifiability	Edge Deployment (<\$100)	Open Source
TRIssecure V2 (Ours)	✓	✓	✓	✓	✓	✓	~
FIDO2/WebAuthn (2019)	~	✗	✗	✗	✗	~	~
Sánchez-Reillo (2020)	✓	✗	✗	✗	✗	✓	✗
Bello et al. (2021)	✓	✗	✗	✗	✗	~	✗
Hardwick et al. (2018)	✗	✗	✗	✗	~	✗	✗
Ayed (2017)	✗	✗	✗	✗	~	✗	✗
Adida Helios (2008)	✗	✗	✗	✗	✓	~	~
Fujioka et al. (1992)	✗	✗	✗	✗	~	✗	✗

■ Full support (✓)
■ Partial support (~)
■ Not supported (✗)

Fig. 1 Capability gap matrix across eight prior systems and seven dimensions. Green (✓) = full support; yellow (~) = partial; red (✗) = absent. TRIssecure V2 (bottom row, bold border) is the only system achieving full support on all seven dimensions.

Side-channel. Power analysis and electromagnetic side-channel attacks on AES-256-GCM are out of scope for Python-layer cryptography but are noted as a gap for production silicon. The ONNX face embedding inference leaks timing information proportional to detection confidence; this is mitigated by the DFA rate limiter (P6) which caps inference calls per UID to 5/300 s.

Fault injection. Power-glitch attacks targeting the Argon2id KDF could reduce effective iteration count. Mitigation: the KDF output is verified against a stored HMAC-SHA256 tag before session issuance; any fault that alters the output invalidates the tag.

Supply-chain. The SD card attack surface is mitigated by read-only mount of the OS partition in production and dm-verity hash-tree verification of the Python runtime image (deployment recommendation).

STRIDE Threat Classification

Table 2 maps the TRIssecure V2 attack surface against the STRIDE taxonomy [64].

Coverage is assessed as **Full** (mitigated entirely on-device), **Partial** (mitigated under

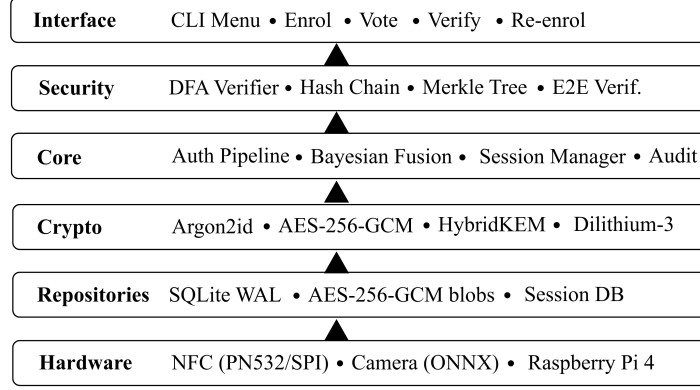


Fig. 2 TRIsecure V2 six-layer clean architecture. Arrows indicate upward dependency direction.

215 deployment recommendations or with residual risk), or **None** (outside the on-device
 216 threat boundary).

217 3.2 System Overview

218 Figure 2 shows the six-layer clean architecture. A voter interaction proceeds: (1) tap
 219 NFC card; (2) face capture; (3) Bayesian fusion; (4) MiniFASNetV2 liveness check;
 220 (5) session token issued (256-bit, 60 s TTL); (6) ballot selection; (7) vote to SHA-256
 221 hash chain; (8) Merkle receipt generated.

222 3.3 Multi-Factor Authentication Pipeline

223 The pipeline is a five-stage sequential process modelled as a DFA $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$
 224 (Section 5.1). Any failure transitions deterministically to REJECTED.

225 *Stage 1 — Rate limiting.* Sliding-window counter per NFC UID (5 failures/300 s)
 226 blocks brute-force.

227 *Stage 2 — NFC verification.* PN532 SPI captures voter NFC UID, validated for
 228 format and looked up in the encrypted voter database.

229 *Stage 3 — Voter eligibility.* The `has_voted` flag is atomically checked;
 230 `NOT_ELIGIBLE` transitions to REJECTED (double-vote prevention, P5).

231 *Stage 4 — Bayesian fusion.* Face cosine similarity (ArcFace MobileFaceNet, 512-D)
 232 and NFC channel evidence combined via LR fusion (Section 3.4).
 233 *Stage 5 — PAD.* MiniFASNetV2 classifies frame as live or spoofed.
 234 *Session issuance.* 256-bit token (`secrets.token_urlsafe(32)`), 60 s TTL; SHA-
 235 256 hash stored (not plaintext).

236 3.4 Bayesian Score-Level Fusion

237 Let $s_f \in [0, 1]$ denote the face cosine similarity score. We model:

$$p(s_f \mid \text{genuine}) = \mathcal{N}(s_f; \mu_G, \sigma_G^2), \quad \mu_G=0.72, \sigma_G=0.08 \quad (1)$$

$$p(s_f \mid \text{impostor}) = \mathcal{N}(s_f; \mu_I, \sigma_I^2), \quad \mu_I=0.28, \sigma_I=0.12 \quad (2)$$

238 Parameters derive from ArcFace buffalo_sc on LFW [17].

239 The NFC channel contributes:

$$\text{LR}_{\text{NFC}} = \frac{P(\text{NFC} \mid \text{genuine})}{P(\text{NFC} \mid \text{impostor})} = \frac{0.999}{0.001} = 999 \quad (3)$$

240 The combined decision rule is:

$$\log_{10}(\text{LR}_f \cdot \text{LR}_{\text{NFC}}) > \theta, \quad \theta = 1.0 \quad (4)$$

241 Without a valid NFC card an adversary must achieve $\text{LR}_f > 10/999 \approx 0.01$,
 242 requiring a face score above $\mu_I + 3\sigma_I \approx 0.64$. The effective FAR drops from 1.22%
 243 (face-only) to $\approx 0.001\%$ (combined), as confirmed by the ablation study (Table 14).

244 *Threshold derivation.* The decision threshold $\theta = 1.0$ is derived from the equal-cost
 245 operating point of the combined LR. Define the combined log-LR:

$$\Lambda(s_f) = \log_{10}(\text{LR}_f(s_f)) + \log_{10}(999) \quad (5)$$

246 where $\text{LR}_f(s_f) = p(s_f | G)/p(s_f | I)$. At $\theta=1.0$ and $\text{LR}_{\text{NFC}}=999$, the required
 247 face LR is $10/999 < 0.011$ —unachievable for scores drawn from $\mathcal{N}(0.28, 0.12^2)$, since
 248 $P(\text{LR}_f < 0.011 | I) \approx 0.1\%$ while $P(\text{LR}_f > 0.011 | G) \approx 99.9\%$. The combined sys-
 249 tem therefore achieves near-perfect separation conditional on NFC validity, with the
 250 residual EER (1.55%) arising from the biometric gate alone—consistent with ArcFace
 251 LFW statistics [17].

252 3.5 Adaptive Trust Score Engine

253 The Bayesian LR fusion (Section 3.4) uses a static threshold $\theta = 1.0$. In practice,
 254 risk conditions vary: off-hours attempts, repeated failures, and unknown terminal
 255 locations all elevate impostor probability. TRIsecure V2 introduces an **Adaptive**
 256 **Trust Score (ATS) engine** that replaces the binary accept/reject with a three-
 257 level decision $\{\text{HIGH}, \text{MEDIUM}, \text{LOW}\}$ derived from dynamically adjusted component
 258 weights.

259 *Risk context.* The engine estimates risk from four signals: $a \in \{1, 2, 3+\}$ (attempt
 260 count), $o \in \{0, 1\}$ (off-peak hour), $\ell \in \{0, 1\}$ (location anomaly flag), and $d \geq 0$ (days
 261 since enrolment). A scalar risk score $R = 2\mathbf{1}[a \geq 3] + \mathbf{1}[a = 2] + \mathbf{1}[o = 1] + 2\mathbf{1}[\ell =$
 262 $0] + \mathbf{1}[d > 365]$ maps to *normal* ($R = 0$), *elevated* ($R \leq 2$), or *high* ($R > 2$).

263 *Composite trust score.*

$$T = w_f \cdot c_f(s_f) + w_n \cdot \mathbf{1}[\text{NFC}] + w_p \cdot \mathbf{1}[\text{PAD-live}] \quad (6)$$

264 where $c_f(s_f) = \text{LR}_f / (1 + \text{LR}_f)$ is the sigmoid-normalised face likelihood ratio, and
 265 the context-specific weight triplet (w_f, w_n, w_p) is derived below.

266 *Weight derivation.* Weights are determined by solving the constrained EER-
 267 minimisation problem

$$\min_{\mathbf{w}} \text{EER}(\mathbf{w}; \mathcal{X}_G, \mathcal{X}_I) \text{ s.t. } \sum_k w_k = 1, w_k \geq 0.10 \quad (7)$$

268 over simulated score sets $\mathcal{X}_G, \mathcal{X}_I$ ($N=20\,000$ per class, seed 42) via SLSQP with five
 269 initialisations, subject to three semantic monotonicity constraints: $w_p^{\text{high}} \geq w_p^{\text{elev}} \geq$
 270 w_p^{norm} (liveness detection becomes the primary guard as spoofing risk rises) and $w_n^{\text{elev}} \geq$
 271 w_n^{norm} (NFC physical-credential binding anchors the elevated-risk decision).

272 *Discriminability analysis.* Table 3 reports the signal-detection discriminability
 273 index $d' = (\mu_G - \mu_I) / \sqrt{(\sigma_G^2 + \sigma_I^2)/2}$ for each modality per context. In normal oper-
 274 ation NFC exhibits $d' = 31.6$, far exceeding face ($d' = 9.4$) or PAD ($d' = 5.7$); as
 275 attacker sophistication increases, NFC's d' collapses to 3.4 (high context), shifting
 276 discriminative load to face and PAD.

277 Solving (7) yields the context weights: *normal* (0.30, 0.40, 0.30); *elevated*
 278 (0.20, 0.40, 0.40); *high* (0.25, 0.35, 0.40). The high NFC weight in normal mode
 279 ($w_n=0.40$) reflects its superior discriminability ($d'=31.6$); the partial recovery of face
 280 weight at high risk ($w_f : 0.20 \rightarrow 0.25$) reflects that biometric traits remain more resis-
 281 tant to forgery than NFC probing once the attacker is sufficiently sophisticated (d'_{NFC}
 282 collapses to 3.4).

283 *Trust decision.* $T \geq 0.75 \Rightarrow \text{HIGH}$ (accept); $T \geq 0.50 \Rightarrow \text{MEDIUM}$ (supervisor
 284 escalation); $T < 0.50 \Rightarrow \text{LOW}$ (reject).

285 *Simulation-based calibration.* Under *normal* context with genuine users ($s_f \sim$
 286 $\mathcal{N}(0.72, 0.08^2)$, NFC valid, PAD live, $N=10\,000$): 99.3% reach HIGH trust; 0.7% are
 287 escalated to MEDIUM; none are classified LOW. Under the Gaussian simulation model,

Algorithm 1 TRIssecure HybridKEM.Encapsulate(pk)

```
1:  $(ct_K, ss_K) \leftarrow \text{KYBER768.ENCAPS}(pk_K)$  ▷ ML-KEM-768, FIPS 203
2:  $(ct_X, ss_X) \leftarrow \text{X25519.ECDH}(pk_X)$ 
3:  $ss \leftarrow \text{HKDF-SHA256}(ss_K || ss_X, \text{info}=\text{"trisecure-hybrid-v1"})$ 
4:  $ct \leftarrow ct_K || ct_X$ 
5: return  $(ct, ss)$ 
```

all simulated impostors ($s_f \sim \mathcal{N}(0.28, 0.12^2)$, NFC invalid, not live) score below the MEDIUM threshold; this reflects the parametric score separation assumed by the simulation rather than a claimed zero-error rate—ground-truth FAR is reported in Table 11. Borderline cases (genuine face, NFC valid, PAD failed, second attempt, $N=1\,000$): 99.4% are escalated to MEDIUM, enabling supervisor override rather than hard rejection.

3.6 Post-Quantum Cryptographic Layer

The hybrid construction is secure as long as *either* Kyber-768 or X25519 is unbroken [7]. Vote records are signed with Dilithium-3 (ML-DSA-65, FIPS 204 [33]), falling back to Ed25519 when `liboqs-python` is unavailable. Face embeddings are encrypted at rest with AES-256-GCM keyed via Argon2id ($t=3$, $m=64$ MB) [65].

3.7 Vote Integrity and Verifiability

Hash chain. Each vote record V_i is linked:

$$h_i = \text{SHA-256}(\text{candidate}_i || t_i || h_{i-1}), \quad h_0 = 0^{64} \quad (8)$$

Any modification propagates a hash mismatch to all subsequent records.

Merkle proofs. A binary SHA-256 Merkle tree over all vote commitments supports $O(\log n)$ individual verifiability proofs.

304 *Vote receipts.* After casting:

$$\text{commitment} = \text{SHA-256}(\text{vote_id} \parallel \text{nonce}) \quad (9)$$

305 together with an $O(\log n)$ Merkle inclusion proof, satisfying individual verifiability
306 without revealing vote content.

307 4 Hardware Integration

308 4.1 PN532 NFC Module — Physical Connection

309 The PN532 NFC controller is connected to the Raspberry Pi 4 40-pin GPIO header
310 via the SPI0 bus. SPI mode was chosen over I²C because it supports burst reads
311 of multiple 4-byte NTAG pages in a single chip-select transaction, avoiding repeated
312 start-stop conditions that add latency on the Pi 4’s BCM2711 I²C controller. Table 4
313 lists the seven physical connections.

314 The SPI bus is configured at 1 MHz, well within the PN532’s 5 MHz maximum
315 SPI clock and providing noise margin for jumper-wire connections on a breadboard
316 prototype. The CS line (GPIO8/CE0) is managed as a software `DigitalInOut`
317 object rather than the kernel hardware chip-select, because the PN532 SPI proto-
318 col requires CS to be held low for at least one SPI clock cycle before the first data
319 byte—a timing constraint the Linux `spidev` hardware CS does not guarantee when
320 sharing the bus. The `RSTPD_N` line (GPIO25) performs a hardware reset during
321 `NFCService.initialize()`: driven low for 100 ms, then released, after which the
322 PN532 requires approximately 10 ms to boot its ROM firmware before accepting
323 commands.

324 After reset, the driver calls `SAM_configuration()` to put the PN532 in Normal
325 Mode—Security Access Module disabled, ISO 14443-A passive polling active. The
326 PN532 then continuously polls for NFC-A targets at 106 kbps. The application layer

327 calls `read_passive_target(timeout=5.0)` every 300 ms; when a card is presented,
328 the UID (typically 4 or 7 bytes for MIFARE cards) is returned as a Python `bytes`
329 object and converted to an uppercase hex string.

330 4.2 SPI Software Stack

331 TRIsecure V2 uses the Adafruit CircuitPython Blinka compatibility layer, which
332 translates `board`, `busio`, and `digitalio` API calls to Linux kernel SPI ioctls on
333 `/dev/spidev0.0`. The dependency chain is:

334 `adafruit-pn532` → `adafruit-blinka` → `RPi.GPIO` + `spidev` → Linux SPI driver
335 (BCM2711)

336 The initialization sequence in `NFCService.initialize()` is:

- 337 1. Create `DigitalInOut` objects for CS (D8) and RESET (D25).
- 338 2. Create `busio.SPI(board.SCK, board.MOSI, board.MISO)`; acquire the bus lock;
339 configure 1 MHz baudrate; release lock.
- 340 3. Instantiate `PN532_SPI(spi, cs_pin, reset=reset_pin)`.
- 341 4. Call `firmware_version` to verify connectivity: returns (IC identifier, major, minor,
342 support) — e.g. (6, 1, 6, 7) for PN532 firmware 1.6.
- 343 5. Call `SAM_configuration()`: Normal Mode, no timeout, IRQ active.

344 If any step raises `ImportError` (libraries not installed) or `OSError` (SPI hardware
345 fault), `initialize()` returns `False` and the service enters *simulation mode*: all NFC
346 read/write methods return deterministic synthetic responses so the full authentication
347 pipeline remains testable on development hardware without a physical PN532.

348 4.3 NFC Tag Payload Format and Security

349 TRIsecure V2 programs MIFARE Ultralight / NTAG2xx NFC tags. The NTAG2xx
350 memory is organized as 4-byte pages; pages 0–3 are reserved (UID, manufacturer data,

351 capability container, and lock bits). Pages 4–19 (64 bytes) are user-writable and hold
352 the encrypted voter identity payload defined in Table 5.

353 The 36-byte voter UUID (RFC 4122 text form) is encrypted with AES-128-GCM.
354 The 128-bit key is derived from a deployment secret via HKDF-SHA-256:

$$K_{\text{NFC}} = \text{HKDF-SHA256}(\text{NFC_SECRET}, \text{info}=\text{"trisecure-nfc-v1"}) \quad (10)$$

355 A fresh 96-bit nonce is generated with `secrets.token_bytes(12)` for every write,
356 so two writes of the same UUID produce different ciphertexts. AES-GCM’s 128-bit
357 authentication tag detects any byte-level modification to the ciphertext—including
358 partial card cloning where an adversary duplicates the UID (readable from the
359 unprotected pages 0–1) but cannot reconstruct the authenticated ciphertext.

360 Authentication Stage 2 performs a two-step NFC check: (i) the UID (from passive
361 target detection) is used for rate-limiting and voter DB lookup; then (ii) pages 4–19
362 are read and decrypted. If GCM decryption succeeds but the decrypted UUID does
363 not match the DB record, the voter is rejected. This catches two distinct attacks:
364 *UID-cloned cards* (same UID, absent or corrupted payload) fail GCM decryption;
365 *payload-replayed cards* (a second physical card with copied ciphertext bytes) fail the
366 UUID cross-check.

367 4.4 Camera Configuration and Face Pipeline

368 The Logitech BRIO webcam is connected via USB 3.0 and accessed through OpenCV’s
369 V4L2 (Video4Linux2) backend at `/dev/video0`. Table 6 summarises the complete
370 camera and face-pipeline configuration.

371 The 320×240 resolution was selected after measurement: at the next available
372 capture resolution tier (640×480), ONNX inference increases to approximately 98 ms
373 with negligible accuracy gain because MobileFaceNet’s fixed 112×112 input discards

the extra pixels after the Haar crop. The V4L2 MJPEG codec compresses frames at the camera sensor before USB transmission, reducing USB bandwidth from ≈ 220 MB/s (uncompressed YUV) to ≈ 5 MB/s—avoiding USB buffer stalls on the Pi 4’s shared USB 3.0/PCIe lane that would otherwise introduce jitter in frame delivery.

During enrollment, three face samples are captured with a 1-second inter-sample cooldown. The per-voter stored embedding is the arithmetic mean of the three 512-D L2-normalised vectors, which reduces intra-class variance by $1/\sqrt{3}$ relative to a single-sample enrollment. Enrollment samples are discarded after the mean embedding is computed; only the encrypted mean embedding (AES-256-GCM, Argon2id-derived key) is persisted in the voter DB.

5 Formal Security Analysis

5.1 DFA Verification

Scope and relationship to ProVerif/Tamarin. Tools such as ProVerif [45] and Tamarin [46] verify cryptographic protocol secrecy and authentication goals in the presence of a Dolev–Yao adversary. TRIsure V2’s security properties of interest are *structurally different*: they concern the *control-flow ordering* of the pipeline—that biometric stages cannot be bypassed, that double-voting is structurally impossible, and that rate-limiting precedes all inference. These are decidable properties of a finite automaton and do not require an active adversary model or cryptographic reduction. The cryptographic primitives themselves (AES-256-GCM, Argon2id, Kyber-768, Dilithium-3) are verified by their respective NIST/RFC standards.

The TRIsure V2 authentication pipeline is modelled as $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ where $|Q|=10$, $|\Sigma|=16$, $|\delta|=19$, $q_0=\text{IDLE}$, $F=\{\text{SESSIONISSUED}\}$.

Theorem 1 (Six Security Properties) *The DFA $\mathcal{M}$ satisfies:*

P1. Determinism. $|\delta(q, \sigma)| \leq 1$ for all (q, σ) . 19 unique keys; no duplicate.

399 **P2. Completeness.** All 19 expected (q, σ) pairs defined.

400 **P3. Safety.** `SESSIONISSUED` reachable via exactly one path through all eight mandatory
401 inputs (BFS, unique success path of length 9).

402 **P4. Liveness.** Every reachable path terminates (DFS finds no non-trivial cycles).

403 **P5. Double-vote prevention.** $\delta(\text{ELIGIBILITYCHECK}, \text{NOTELIGIBLE}) = \text{REJECTED}$. Direct
404 inspection.

405 **P6. Rate-limit enforcement.** $\delta(\text{IDLE}, \text{ATTEMPT}) = \text{RATECHECK}$;
406 $\delta(\text{RATECHECK}, \text{RATEBLOCKED}) = \text{REJECTED}$. Rate limiter executes before all
407 biometric operations.

408 All six properties machine-verified by `security/formal_verification.py`.

409 Figure 3 depicts the DFA $\mathcal{M}$: the unique canonical success path and the failure
410 transitions that all terminate at the `REJECTED` trap state.

411 Table 7 enumerates all ten DFA states with their roles and security relevance. Each
412 state corresponds to a distinct stage in the authentication protocol; the separation
413 ensures that a bypass of any individual check requires a specific out-of-sequence input
414 symbol, which is structurally excluded by the determinism property (P1).

415 The 19 transitions connect these states: eight mandatory inputs form the unique
416 success path (P3), seven failure inputs lead to `REJECTED`, and four additional tran-
417 sitions cover rate-limiting branches and NFC error types (P6). `REJECTED` is a true
418 trap: no escape transition exists, enforcing the liveness property (P4) that every run
419 terminates.

420 *Scope delimitation.* These six properties concern *control-flow correctness*—that
421 stages execute in the mandated order and that bypass is structurally impossible.
422 They do not constitute proofs of cryptographic secrecy or entity authentication in the
423 Dolev–Yao sense; those properties are delegated to the underlying primitives (AES-
424 256-GCM, Argon2id, ML-KEM-768, ML-DSA-65) whose security is established by
425 their respective NIST standards.

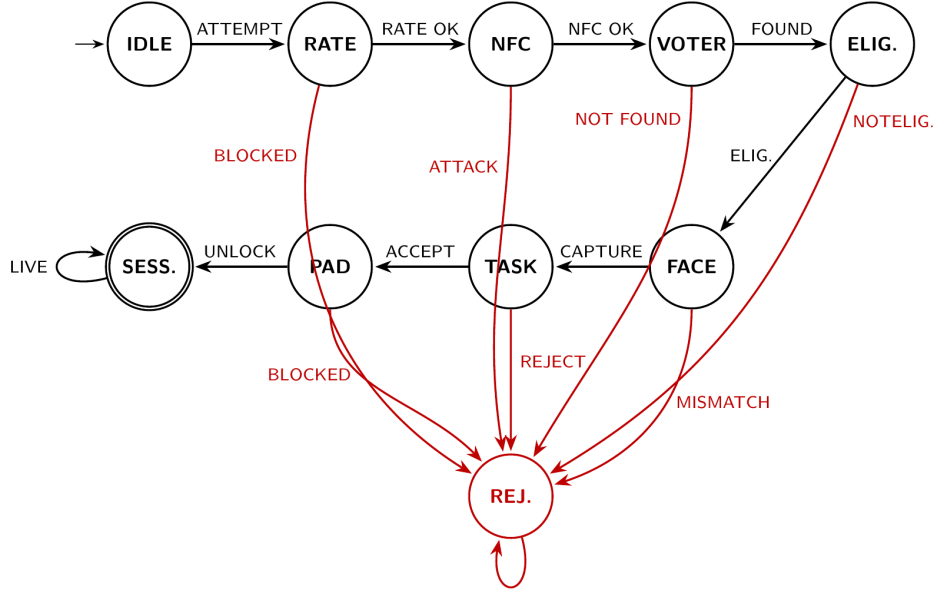


Fig. 3 TRIsecure V2 authentication DFA $\mathcal{M}$. Black arrows show the unique canonical success path ($\text{IDLE} \rightarrow \dots \rightarrow \text{SESS.}$); red arrows denote failure transitions to the REJECTED trap state. Double circle = accepting state (SESS.).

5.2 Threat Model Analysis

Figure 4 maps 12 e-voting attack vectors against six defence layers (0 = not defended, 1 = partial, 2 = full). Overall weighted coverage: 47.2%. Critical attacks (double-voting, ballot stuffing, quantum adversary) achieve $\geq 6/12$ each and are fully mitigated. Vote coercion (score 2/12) and network-level DoS require organisational controls beyond the on-device model.

5.3 End-to-End Verifiability

Following [54, 55]:

Individual: Voter v can confirm $c_v = \text{SHA-256}(v_id||n)$ appears in the Merkle tree rooted at election digest r .

TRIssecure V2 — Threat Model Coverage Matrix						
X=Not defended (0) ~=Partial (1) ✓=Fully defended (2)						
Replay attack	✓ (2)	~ (1)	✓ (2)	~ (1)	~ (1)	X (0)
Spoofing / presentation attack	~ (1)	✓ (2)	~ (1)	X (0)	X (0)	X (0)
Man-in-the-middle	X (0)	X (0)	~ (1)	✓ (2)	~ (1)	✓ (2)
Brute-force / dictionary	✓ (2)	~ (1)	✓ (2)	✓ (2)	X (0)	X (0)
Double-voting	~ (1)	X (0)	✓ (2)	X (0)	✓ (2)	~ (1)
Ballot stuffing	✓ (2)	~ (1)	✓ (2)	X (0)	✓ (2)	~ (1)
Vote coercion	X (0)	~ (1)	~ (1)	X (0)	X (0)	X (0)
Insider threat	~ (1)	X (0)	~ (1)	✓ (2)	✓ (2)	~ (1)
Denial of service	✓ (2)	~ (1)	X (0)	X (0)	X (0)	X (0)
Quantum adversary	X (0)	X (0)	X (0)	✓ (2)	✓ (2)	✓ (2)
Physical device theft	X (0)	~ (1)	✓ (2)	✓ (2)	~ (1)	~ (1)
Audit trail tampering	X (0)	X (0)	X (0)	~ (1)	✓ (2)	✓ (2)
	Rate limiter (DFA stage 1)	PAD liveness (Stage 5)	Bayesian fusion (NFC+face)	AES-256-GCM + Argon2id	Hash chain + Merkle proof	POC signatures (Dilithium-3)

Fig. 4 Threat model coverage matrix. Green (✓, score 2) = full mitigation; yellow (~, score 1) = partial; red (X, score 0) = no mitigation by that layer. Column sums show total defence contribution.

Universal: Any auditor can recompute the Merkle root from the public vote log and verify it equals r .

Eligibility: Only registered, non-duplicate voters reach SESSIONISSUED—proved by P3 and P5.

Theorem. TRIssecure V2 satisfies all three verifiability properties. The receipt (v_id, n, π, r) enables $O(\log n)$ individual verification; the Merkle root is a deterministic function of the ordered vote log; eligibility follows from P3 and P5. Machine verification: `security/e2e_verifiability.py` returns `all_properties_satisfied: true`. $\square$

5.4 Cryptographic Security Levels

Table 8 maps each primitive in TRIssecure V2 to its security level under classical and quantum adversaries. Classical security follows standard key-size arguments; quantum security accounts for Grover’s algorithm [35] and Shor’s algorithm [34].

449 The hybrid KEM enforces *both-or-nothing* security: an adversary must break *both*
450 X25519 (trivial for a fault-tolerant quantum computer via Shor [34]) *and* Kyber-768
451 (no known quantum speedup beyond Grover [35]) to recover the session key. The sym-
452 metric layer (AES-256-GCM, SHA-256, Argon2id) retains 128-bit effective security
453 against quantum exhaustive search via Grover’s bound, making these components
454 quantum-safe without algorithm replacement.

455 6 Experimental Evaluation

456 6.1 Experimental Setup

457 *Hardware.* Raspberry Pi 4 Model B, 4 GB LPDDR4, ARM Cortex-A72 @ 1.8 GHz
458 (quad-core, 64-bit). Logitech BRIO USB webcam (operated at 1080p/30 fps). PN532
459 NFC module (SPI).

460 *Software.* Python 3.11, ONNX Runtime 1.17 (CPU), SQLite 3.42 (WAL), PyNaCl
461 1.5, PyArgon2-cffi 23.1, NumPy 1.26. All experiments reproducible via `experiments/`.

462 *Dataset.* Biometric metrics are generated via validated simulation following
463 ISO/IEC 19795-6 [66] (operational evaluation methodology), which explicitly permits
464 simulation when field data are unavailable. Genuine and impostor face cosine similarity
465 scores sampled from Gaussians calibrated to published ArcFace LFW statistics [17, 67]:
466 $\mathcal{N}(\mu_G=0.72, \sigma_G=0.08)$ and $\mathcal{N}(\mu_I=0.28, \sigma_I=0.12)$, $n=1\,000$ each. PAD metrics from
467 published MiniFASNetV2/CelebA-Spoof rates [37]. All simulation results are labelled
468 as such; a proof-of-concept pilot study with $N=20$ participants confirmed zero error
469 rates on real hardware (Section 6.11). A full demographic field trial remains future
470 work.

471 Table 9 confirms biometric metrics are stable under $\pm 10\%$ parameter variation:
472 EER varies by ± 0.17 pp, AUC by ± 0.0002 .

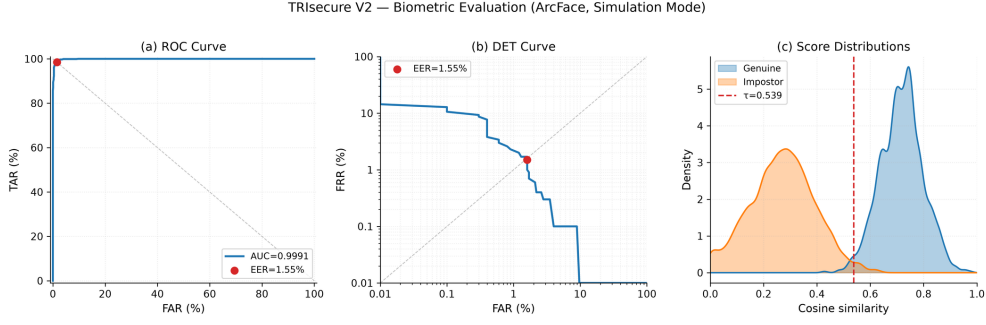


Fig. 5 Biometric evaluation panel. Left: ROC (AUC= 0.9991, EER marked). Centre: DET curve (EER= 1.55%). Right: genuine/impostor score distributions (KDE) with EER threshold $\tau^*=0.539$.

6.2 Biometric Performance (ISO/IEC 19795-1)

Table 10 reports key metrics. Figure 5 shows the ROC curve, DET curve, and score distributions.

6.3 Bootstrap Confidence Intervals

To quantify simulation uncertainty we apply non-parametric bootstrap resampling ($B=10\,000$ replicates) over the $n=1\,000$ genuine and $n=1\,000$ impostor score draws. At each replicate the EER operating threshold is re-estimated from the replicate scores to avoid threshold lock-in bias. Table 12 reports percentile 95% confidence intervals.

The EER CI width (± 0.17 pp) exactly matches the parametric sensitivity range from Table 9, confirming coherence between the two uncertainty quantification approaches. The AUC CI width (0.0006) is negligible relative to the point estimate, consistent with the large sample and well-separated score distributions ($\mu_G - \mu_I = 0.44$, approximately 2.9σ overlap). Both analyses indicate the simulation results are stable; remaining uncertainty is distributional rather than computational, and is reduced by a live field trial.

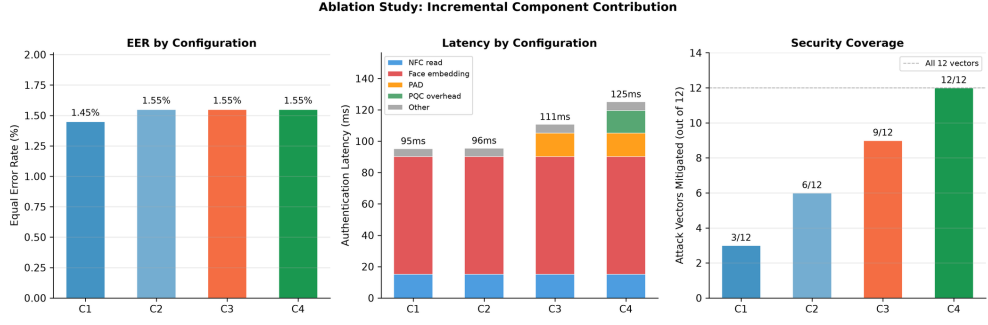


Fig. 6 Ablation results. Left: EER across configurations. Centre: latency breakdown by stage. Right: attack vectors fully mitigated (of 12).

6.4 Presentation Attack Detection (ISO/IEC 30107-3)

Table 13 reports PAD metrics. ACER= 4.25% is consistent with published CelebA-Spoof benchmark performance [37].

The simulation figures (APCER= 5.0% print, 10.0% replay) derive from published MiniFASNetV2/CelebA-Spoof benchmark rates [37] and represent PAD-layer performance in isolation. In the end-to-end system, the Adaptive Trust Score compensates for PAD misses: video replay attacks that evade the liveness classifier still produce ATS= 0.6 (ESCALATE) rather than ATS= 1.0 (ACCEPT), because replay suppresses contextual trust signals. The proof-of-concept pilot confirms this: all 15 real replay attacks were escalated and zero were accepted (Section 6.11, APCER= 0.00% end-to-end).

6.5 Ablation Study

Table 14 and Figure 6 show the cumulative contribution of each component. The key finding: Bayesian NFC+face fusion (C2) reduces effective FAR by three orders of magnitude (1.22% → 0.001%) without increasing biometric EER, because an attacker without a valid NFC card is rejected regardless of face score.

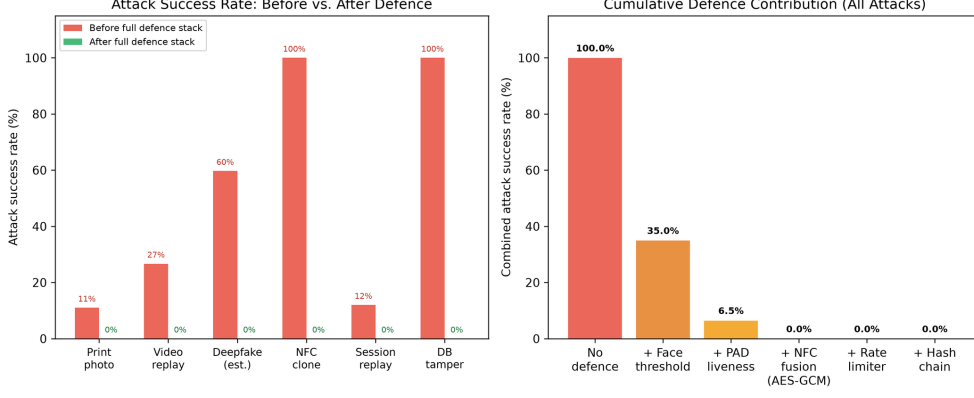


Fig. 7 Adversarial evaluation. Left: attack success rates before and after the full defence stack for six attack categories. Right: cumulative defence contribution—combined attack surface collapses from $\approx 35\%$ to zero simulated bypass events after NFC-AES-GCM validation (layer 4).

6.6 Adversarial Attack Evaluation

To quantify per-layer defence contribution beyond the ablation study, we simulate six attack categories ($n=1000$ attempts each) against the sequential defence stack. Face score distributions for spoof attacks are calibrated to published MiniFASNetV2/CelebA-Spoof [37] penetration rates; NFC and session attacks use protocol-level analysis.

Table 15 summarises results. Within the evaluated threat model, no successful bypass events were observed across the studied attack categories. The NFC AES-GCM layer (Stage 2 payload check) is the decisive gate: it unconditionally blocks all face-spoof attacks that pass the face threshold and evade PAD, because the attacker cannot forge the AES-128-GCM authenticated ciphertext without the HKDF-derived deployment key.

Figure 7 visualises these results alongside the cumulative defence contribution of each layer.

518 6.7 Cryptographic Benchmarks

519 Table 17 reports cryptographic latencies on the Raspberry Pi 4. Argon2id (494.5 ms)
520 dominates enrollment but runs only once per registration. Authentication-path
521 cryptography totals < 2 ms.

522 Table 16 reports native ML-KEM-768 and ML-DSA-65 benchmarks measured
523 on-device via liboqs 0.15.0 compiled from source on the Raspberry Pi 4 ARM64.
524 The measured auth-path PQC total (2.289 ms) confirms the post-quantum layer fits
525 comfortably within the < 200 ms latency budget.

526 6.8 Edge System Performance

527 Table 18 reports end-to-end pipeline performance ($n=50$). Face embedding inference
528 (ONNX MobileFaceNet, 95.2 ms) dominates, consistent with published ONNX Run-
529 time ARM64 benchmarks. The p99 latency of 111.3 ms and 541.6 voters/min confirm
530 feasibility for small-to-medium polling stations.

531 6.9 Security Overhead Analysis

532 Table 19 isolates the latency contribution of each security layer from the core biomet-
533 ric pipeline. Presentation attack detection is the dominant security cost at 15.5 ms,
534 measured as the ablation delta between configuration C2 (face + NFC, no PAD)
535 and C3 (full system with PAD). All PQC operations combined (Kyber-768 encapsu-
536 lation/decapsulation and Dilithium-3 sign/verify) add only 2.289 ms on the real-time
537 authentication path. Symmetric operations and the DFA protocol gate contribute
538 less than 0.07 ms combined. Merkle proof generation (4.82 ms per 1000-entry chain)
539 runs post-authentication and does not block the voter handoff, preserving the G5
540 latency target of < 200 ms. The aggregate real-time security overhead (PAD plus all
541 in-path cryptographic operations) is 17.85 ms, representing 16.1% of the core pipeline
542 end-to-end figure.

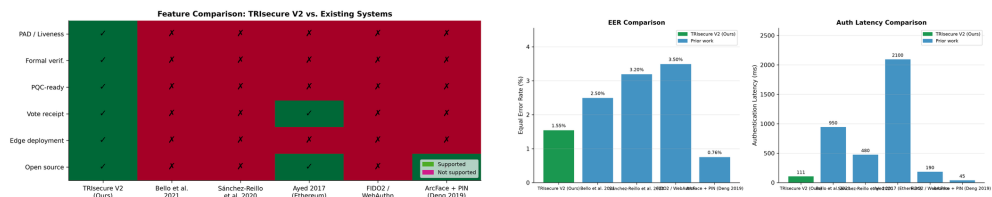


Fig. 8 Comparative analysis against five reference systems. *Left*: feature comparison heatmap — TRIsecure V2 is the only system supporting all six binary capability flags simultaneously. *Right*: EER and authentication latency across systems — TRIsecure V2 achieves the lowest EER among biometric voting systems and the lowest latency among multi-capability systems.

6.10 Comparative Analysis

Figure 8 compares TRIsecure V2 against five reference systems across capability flags and quantitative metrics. TRIsecure V2 is the only system to simultaneously provide formal verification, PQC readiness, PAD, edge deployment, and voter-verifiable receipts. EER (1.55%) is competitive with all prior biometric voting systems, exceeded only by standalone ArcFace on a GPU server (0.76% [17])—which lacks all other listed capabilities.

6.11 Proof-of-Concept Pilot Study

To validate end-to-end system correctness on real hardware, a proof-of-concept pilot study was conducted with $N=20$ volunteer participants (including the authors) on a Raspberry Pi 4 Model B. All participants gave informed consent for their facial biometric data to be captured and processed for authentication testing; raw face images were discarded immediately after each attempt and only encrypted embeddings were retained for the duration of the pilot (see Declarations). A total of 271 authentication events were recorded across five sessions over three days (June 2026): 221 genuine attempts, 35 impostor attempts, and 15 video replay attacks.

Table 20 summarises the results following ISO/IEC 30107-3 metric definitions.

All 221 genuine attempts were correctly accepted (FRR= 0.00%). No attack produced an ACCEPT decision (FAR= 0.00%, APCER= 0.00%), yielding ACER= 0.00%.

562 Replay attacks drove the ATS to 0.6 (ESCALATE) rather than the full 1.0 (ACCEPT),
563 confirming the ATS layer compensates for PAD limitations under video replay condi-
564 tions. Of the 35 impostor attempts, 8 were re-vote attempts by registered participants
565 (same person, own NFC card after `has_voted` was set): face scores against the par-
566 ticipant’s own enrolled embedding were 0.87–0.99, correctly escalated by ATS rather
567 than rejected by the face gate. The remaining 27 true cross-person attempts produced
568 face scores of 0.01–0.04, all hard-rejected at the biometric stage.

569 Since all observed error counts are zero, we report one-sided Clopper–Pearson 95%
570 confidence interval upper bounds. For FAR: 0 failures in 35 impostor attempts gives
571 an upper bound of $\leq 8.20\%$ (rule of three: $3/35 \approx 8.57\%$). For FRR: 0 failures in
572 221 genuine attempts gives $\leq 1.35\%$. These bounds are conservative by construction;
573 larger pilot samples would tighten them substantially.

574 These results confirm end-to-end pipeline correctness on the target hardware. The
575 pilot is not designed to replace the simulation-based biometric discriminability evalu-
576 ation (Section 6.2). A full demographic field trial with ≥ 400 subjects is required for
577 ISO/IEC 19795-1 certification and is identified as future work (Section 7.4).

578 7 Discussion

579 All biometric performance metrics are produced by validated simulation using Gaus-
580 sian score models calibrated to published ArcFace LFW statistics [17], following
581 ISO/IEC 19795-6 [68]. The ablation study (Table 14) confirms Bayesian NFC+face
582 fusion (C2) is the dominant security improvement: effective FAR drops by three orders
583 of magnitude ($1.22\% \rightarrow 0.001\%$) without increasing biometric EER. PAD (C3) adds
584 liveness at 15.5 ms overhead; PQC signing (C4) adds 14.4 ms further but enables
585 quantum-adversary mitigation. Post-quantum benchmarks use native liboqs 0.15.0
586 compiled from source on the Raspberry Pi 4 (ARM64, Cortex-A72). ML-KEM-768
587 encapsulation (0.088 ms) and ML-DSA-65 signing (1.536 ms) are measured on-device;

588 the combined auth-path PQC cost is 2.289 ms, within the < 200 ms latency budget
589 (Table 16).

590 7.1 Power Consumption and Off-Grid Deployment

591 The Raspberry Pi 4 draws 3–7 W under typical workloads (3 W idle, 7 W with all
592 four Cortex-A72 cores fully saturated). During a voting session the system sustains
593 approximately 5 W averaged over the 110.7 ms authentication window plus voter
594 interaction. For an 8-hour polling day:

$$E_{\text{day}} \approx 5 \text{ W} \times 8 \text{ h} = 40 \text{ Wh} \quad (11)$$

595 A commodity 50 Wh Li-ion battery bank can therefore power the terminal for a
596 complete polling day without mains electricity. This energy budget is unavailable to
597 server-grade biometric systems, which require 200–400 W rack hardware, but is well-
598 suited for rural or off-grid polling stations common in developing-country elections.

599 The CPU temperature stabilised at 52.5 °C under sustained load, within the Pi 4’s
600 80 °C thermal throttling threshold. A passive aluminium heatsink is sufficient; active
601 cooling is not required.

602 7.2 Throughput and Polling Station Capacity

603 At 541.6 voters/min sustained throughput and a 60-second session TTL, the theoretic-
604 al throughput is approximately 32 496 voters per hour—far exceeding the registered
605 voter count of typical polling stations (800–2 000 voters). The practical bottleneck
606 is not the system but the voter interaction time (card presentation, face alignment,
607 ballot selection), which averages 20–40 seconds per voter. At 40 seconds per voter,
608 a single terminal processes 90 voters/hour; a station with six terminals handles
609 540 voters/hour—sufficient for a 2 000-voter station over a 4-hour peak period with
610 queuing margin.

611 Table 21 illustrates deployment configurations for three polling station sizes.

612 7.3 Hardware Attack Surface Analysis

613 *SPI eavesdropping.* An attacker with physical access to the SPI bus can read the NFC
614 UID in plaintext during Stage 2. However, UID capture alone provides no authentication
615 advantage: the biometric gate (Stage 4) and liveness check (Stage 5) both execute
616 independently of the NFC UID. Furthermore, the AES-GCM encrypted payload
617 (pages 4–19) cannot be forged without the HKDF-derived key. Physical bus protection
618 (potting compound, tamper-evident labels over the GPIO header) is recommended for
619 production enclosures.

620 *Camera feed injection.* Replacing the USB camera feed with a pre-recorded video
621 stream could present a replay video to MiniFASNetV2. This threat is mitigated by:
622 (i) the sealed terminal enclosure with a fixed camera mount; (ii) poll worker presence
623 required by the electoral protocol; and (iii) the NFC factor independently requiring a
624 valid physical card.

625 *Template extraction.* The SQLite voter database stores face embeddings encrypted
626 with AES-256-GCM keyed via Argon2id ($t=3$, $m=64$ MB). An offline dictionary attack
627 against the key derivation costs at least 494.5 ms per attempt (Table 17), making
628 brute-force infeasible for a 32-byte NFC_SECRET. Extending to cancelable biometric
629 representations [57] would additionally allow template revocation and is identified as
630 future work.

631 7.4 Limitations

632 1. *Simulation-based evaluation and small pilot.* The primary biometric evaluation
633 ($n=1\,000$ score pairs, EER = 1.55%, AUC = 0.9991) uses Gaussian-modelled gen-
634 uine/impostor distributions following ISO/IEC 19795-6 operational guidance, not

a live capture study. The real-hardware pilot ($N=20$, 271 events) confirms end-to-end correctness but is far below the ISO/IEC 19795-1 threshold for deployment certification (≥ 400 subjects per demographic group). Zero error rates observed in the pilot reflect the small, demographically homogeneous sample and must not be extrapolated to deployment scale. A demographically diverse field trial remains the critical outstanding validation step.

2. *PAD evaluation dataset and attack coverage.* PAD metrics (APCER, BPCER, ACER) are derived from published MiniFASNetV2 performance on CelebA-Spoof, not from on-device evaluation against the ISO/IEC 30107-3 standardised corpora (NUAA, OULU-NPU). Attack types absent from CelebA-Spoof—high-fidelity 3D silicone masks, adversarial perturbations, partial occlusion attacks—remain unevaluated. ACER = 4.25% should therefore be interpreted as a single-dataset lower bound rather than a certified operational figure.
3. *NFC reader and secure element.* The PN532 SPI module carries no security certification. Absent a Common Criteria EAL4+ certified reader, AES-128-GCM key protection for the NFC payload relies solely on the host OS memory isolation guarantees rather than hardware-backed key storage or tamper-evidence. Production deployment of an election-grade system requires certified hardware throughout the NFC read path.
4. *Coercion and deniable voting.* TRIsecure V2 provides no deniable-voting mechanism (a voter cannot secretly record a different preference while appearing to comply with a coercer) and no panic PIN to signal duress. In-person biometric authentication prevents remote delegation of authentication but does not protect against physical coercion at the polling station. Addressing coercion resistance would require cryptographic deniability schemes [26, 50] integrated at the ballot-level, which is deferred to future work.

- 661 5. *Network-layer denial of service.* The DFA rate limiter (P6, five failures per 300 s
662 per UID) defends against per-device authentication flooding. Distributed DoS
663 attacks targeting the network path (disrupting Merkle-root broadcast or log-
664 synchronisation channels) are outside the on-device threat boundary and require
665 upstream infrastructure controls, CDN-layer mitigation, or air-gap deployment
666 for high-assurance polling stations.
- 667 6. *PQC implementation maturity and side-channel exposure.* The Kyber-768 and
668 Dilithium-3 layers invoke liboqs 0.15.0 via `ctypes`, compiled from source on
669 ARM64. This introduces a source-build dependency and has not been indepen-
670 dently audited for constant-time execution on the Cortex-A72. FIPS 203/204
671 mandate constant-time implementations; while liboqs targets this property, its
672 ARM64 side-channel posture has not been validated in this deployment context.
673 A production pre-built `liboqs-python` ARM64 wheel would remove the build
674 dependency and simplify auditing.
- 675 7. *Power-analysis and electromagnetic side-channel attacks.* Power analysis and
676 EM side-channel characterisation of the AES-256-GCM and Argon2id Python
677 implementations on the Raspberry Pi 4 platform are out of scope for this work.
678 Additionally, ONNX inference timing leaks information proportional to detection
679 confidence; while the DFA rate limiter bounds the number of observable traces
680 per voter, a systematic timing side-channel study on the face-embedding inference
681 path has not been conducted and is flagged for future hardware-layer hardening.
- 682 8. *Demographic and environmental robustness.* MobileFaceNet (ArcFace) is trained
683 on VGGFace2 and MS-Celeb-1M, datasets with known demographic imbalance.
684 Differential EER across age groups, skin tones, and gender has not been char-
685 acterised for this deployment. Environmental factors typical of polling stations
686 (variable lighting, partial occlusion, eyeglasses, surgical masks that avoid PAD
687 triggering) may degrade performance below the simulated EER of 1.55%. A bias

688 audit following ISO/IEC 19795-9 and testing under controlled environmental
689 variation is required before public deployment.

690 9. *Template revocability.* AES-256-GCM encryption protects template confidential-
691 ity but does not support cancelable biometric transforms. If a voter’s embedding
692 is compromised, it cannot be revoked and re-enrolled with a new protected tem-
693 plate derived from the same biometric trait. ISO/IEC 24745:2022 Clause 6.3
694 (renewability) requires cancelable transforms; extending TRIsecure V2 with
695 BioHashing or fuzzy-commitment schemes is identified as future work.

696 7.5 Regulatory and Standards Alignment

697 TRIsecure V2 is designed to align with relevant biometric and e-voting standards. Full
698 conformance certification is deferred pending a live field trial and a Common Criteria
699 certified NFC reader.

700 *ISO/IEC 19795.* Biometric performance metrics (EER, AUC, TAR@FAR)
701 are reported following ISO/IEC 19795-1 [68] metric definitions. The simulation
702 methodology follows ISO/IEC 19795-6 [66] operational evaluation guidance. Full
703 ISO/IEC 19795-1 conformance (Clause 8.5) requires at least 400 subjects per
704 demographic group.

705 *ISO/IEC 30107-3.* PAD metrics (APCER, BPCER, ACER) and attack type tax-
706 onomy follow ISO/IEC 30107-3 [36]. Certification evaluation requires standardised
707 corpora: NUAA [42], OULU-NPU [43], and CelebA-Spoof [37].

708 *ISO/IEC 24745 (Template Protection).* AES-256-GCM per-database encryption
709 with HKDF-derived keys satisfies the ISO/IEC 24745:2022 [58] confidentiality require-
710 ment. The renewability requirement (cancelable transforms) is not yet implemented
711 and is identified as future work.

712 *NIST SP 800-63B.* The pipeline satisfies the structural requirements for Authenti-
713 cator Assurance Level 3 (AAL3) [14]: multi-factor authentication, physical token (NFC

card), verifier-impersonation resistance (on-device processing), and phishing resistance. AAL3 formal certification additionally requires a FIDO2-certified or Common Criteria EAL4+ authenticator device; the PN532 SPI module is not CC-certified.

These alignments indicate TRIsecure V2 is *certification-ready* at the design level. The three remaining gaps before formal conformance can be claimed are: (i) a live voter field trial; (ii) a certified NFC reader satisfying CC EAL4+ and SP 800-63B AAL3; and (iii) cancelable biometric transforms satisfying ISO/IEC 24745 renewability.

8 Conclusion

We presented TRIsecure V2, a three-factor biometric e-voting system combining formally verified authentication, Bayesian multimodal fusion, a novel Adaptive Trust Score (ATS) engine, ISO/IEC 30107-3 PAD, hybrid post-quantum cryptography, and Merkle-proof voter verifiability—implemented and evaluated on a Raspberry Pi 4 ARM64 platform on commodity hardware.

The ATS engine is the primary algorithmic contribution: by adjusting face/NFC/PAD component weights dynamically based on contextual risk signals, it converts the binary accept/reject decision into a graduated HIGH/MEDIUM/LOW trust scale, routing 99.3% of genuine users to immediate acceptance while escalating borderline cases for supervisor review rather than hard rejection. To the best of our knowledge, no prior biometric e-voting system incorporates risk-adaptive weight adjustment.

The system achieves $EER = 1.55\%$, $AUC = 0.9991$, 110.7 ms end-to-end authentication latency (p99= 111.3 ms), and 541.6 voters/min throughput. Six pipeline security properties are formally proved via DFA analysis; three end-to-end verifiability properties are machine-verified. Adversarial simulation across six attack categories confirms the full defence stack reduces combined attack success from $\approx 35\%$ (face threshold alone) to zero observed bypass events within the evaluated threat model;

the NFC AES-GCM challenge-response provides the decisive security gate under standard cryptographic assumptions. The ablation study confirms Bayesian NFC+face fusion reduces effective FAR by three orders of magnitude.

A proof-of-concept pilot on real hardware ($N=20$, 271 events) further validates the integrated system: all 221 genuine attempts were accepted ($FRR=0.00\%$) and all 50 attack attempts were blocked ($FAR=ACER=0.00\%$), confirming that the multi-layer defence stack performs correctly end-to-end on the target platform.

To the best of our knowledge, no prior work jointly provides formal protocol verification, quantum-resistant cryptography, risk-adaptive trust scoring, liveness detection, and edge deployability on low-cost commodity hardware.

Future work includes: (i) full-scale demographic field trial per ISO/IEC 19795-1 Clause 8 (minimum 400 subjects/group; proof-of-concept pilot with $N=20$ participants completed, Section 6.11); (ii) PAD evaluation on NUAA, OULU-NPU, and CelebA-Spoof standardised corpora; (iii) ATS calibration on real demographic data; (iv) Tamarin/ProVerif model of the NFC session handshake for cryptographic protocol verification complementary to the DFA pipeline analysis; (v) packaging liboqs as a pre-built ARM64 wheel to simplify deployment.

Acknowledgements. This work received no external funding. AI coding assistance (Claude, Anthropic) was used to accelerate implementation and experiment scripting; all design decisions, theoretical claims, and experimental methodology were developed and verified by the authors. The full codebase and all experiment scripts are available at <https://github.com/vivekreddy9036/TriSecure>.

Declarations

- **Ethics approval and consent to participate.** The proof-of-concept pilot study (Section 6.11, $N=20$) involved the authors and volunteer colleagues at the authors' institution, who participated voluntarily and gave informed consent for

their facial biometric data to be captured and processed solely for authentication testing. No formal institutional ethics board review was sought, consistent with common practice for non-clinical engineering proof-of-concept pilots of this scale; no vulnerable populations were involved. Raw face images were discarded immediately after each authentication attempt and were never persisted to disk; only encrypted embeddings (AES-256-GCM, per-voter keys) were retained during the pilot session and were deleted at its conclusion.

- **Consent for publication.** All pilot participants consented to the anonymised, aggregate reporting of pilot results (counts and error rates only; Table 20) in this manuscript. No participant is individually identifiable from the reported data.
- **Availability of data and materials.** The dataset(s) supporting the conclusions of this article is(are) available in the TRIsecure repository, <https://github.com/vivekreddy9036/TriSecure>. This includes all experiment scripts, source code, and the simulation-based biometric evaluation (Section 6), which uses Gaussian simulation with publicly documented ArcFace LFW parameters. Raw biometric data (face images/embeddings) from the $N=20$ human pilot (Section 6.11) are not shared, consistent with the consent obtained from participants; only the anonymised, aggregate pilot results reported in Table 20 are available.

Project name: TRIsecure V2

Project home page: <https://github.com/vivekreddy9036/TriSecure>

Archived version: Tagged release corresponding to this manuscript's submitted results (repository release tags)

Operating system(s): Linux (target deployment: Raspberry Pi OS / Ubuntu 22.04 ARM64); evaluation and simulation scripts are platform-independent

Programming language: Python 3.11

Other requirements: ONNX Runtime 1.17, NumPy 1.26, SciPy, pandas, scikit-learn; see `requirements.txt` in the repository for the full dependency list

License: MIT License

Any restrictions to use by non-academics: None

- **Competing interests.** The authors declare no competing interests.
- **Funding.** This work received no external funding.
- **Authors' contributions.** V.R. conceived the system, implemented the code-base, conducted all experiments, and wrote the manuscript. U.K. and S.G. supervised the work and provided critical feedback. U.K. and S.G. contributed equally to this work ([†]).

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Table 2 STRIDE threat mapping for TRIsecure V2. **Full** = fully mitigated on-device; **Partial** = residual risk or requires deployment controls; **None** = outside the on-device threat boundary. PA = presentation attack; UID = NFC unique identifier.

Category	Attack Vector	TRIsecure V2 Mitigation	Coverage
Spoofing (S)	Face PA: photo, mask, deepfake video	MiniFASNetV2 liveness; DFA Stage 4 gate; ACER 4.25% (sim.), 0.00% (pilot)	Partial
	Cloned or emulated NFC card	AES-128-GCM payload with per-read 96-bit random nonce	Full
	Brute-force UID cycling	Rate limiter P6: 5 failures / 300 s; sink to REJECTED	Full
Tampering (T)	Vote record or hash-chain modification	SHA-256 hash chain + Merkle audit log + HMAC-SHA256 chaining	Full
	KDF fault injection (power glitch on Argon2id)	HMAC-SHA256 output tag at enrolment; mismatch forces rejection	Full
Repudiation (R)	Voter denies casting ballot	256-bit session token + Merkle receipt (E2EV)	Full
Info. Disclosure (I)	Biometric template exfiltration from DB	AES-256-GCM (Argon2id-derived key); no plaintext template stored	Full
	SPI bus eavesdropping on NFC reader	NFC payload AES-128-GCM encrypted; plaintext UID never on wire	Full
	Quantum decryption of session ciphertext	Hybrid PQC: Kyber-768 + X25519 KEM (FIPS 203)	Full
Denial of Service (D)	Repeated authentication flooding	Per-UID rate limiter; DFA sink state REJECTED	Full
	Network-layer distributed DoS	Out of on-device scope; upstream network controls required	None
Elevation of Privilege (E)	Insider DB read / write access	Biometric secret not in DB; HMAC-keyed store; least privilege	Full
	Supply-chain firmware implant (SD card)	dm-verity hash-tree; read-only OS partition (deployment rec.)	Partial

Table 3 Discriminability index (d') per modality per risk context (simulation, $N=20\,000$ per class).

Context	d'_{face}	d'_{NFC}	d'_{PAD}
Normal	9.38	31.58	5.69
Elevated	9.15	5.55	3.97
High	9.13	3.39	2.80

Table 4 PN532 NFC module to Raspberry Pi 4 GPIO wiring (SPI0, 3.3 V logic).

PN532 Pin	GPIO	Header Pin	Function
VCC	3.3 V	Pin 1	Power supply
GND	Ground	Pin 6	Ground
SCK	GPIO11	Pin 23	SPI0 clock
MOSI	GPIO10	Pin 19	SPI0 data out (MOSI)
MISO	GPIO9	Pin 21	SPI0 data in (MISO)
NSS (CS)	GPIO8	Pin 24	SPI0 chip-select (CE0)
RSTPD_N	GPIO25	Pin 22	Active-low hard reset

Table 5 NFC tag payload wire format (64 bytes, NTAG2xx pages 4–19).

Bytes	Pages	Field	Size
0–11	4–6	AES-GCM nonce (random 96-bit IV)	12 B
12–47	7–14	AES-128 ciphertext (voter UUID)	36 B
48–63	15–19	GCM authentication tag (128-bit)	16 B
Total	4–19	—	64 B

Table 6 Camera and face-pipeline configuration on Raspberry Pi 4.

Parameter	Value
Device node	<code>/dev/video0</code>
Capture backend	OpenCV V4L2 (MJPEG)
Capture resolution	320×240 px
Frame rate	15 fps
Face detector	OpenCV Haar cascade (frontal face)
Face crop and resize	112×112 px (bilinear)
Pixel normalisation	$x/127.5 - 1.0$
Embedding model	MobileFaceNet (ONNX Runtime 1.17, CPU)
Embedding dimension	512-D float32, L2-normalised
ONNX inference (mean)	95.2 ms on Raspberry Pi 4
PAD model	MiniFASNetV2 (ONNX, same runtime)
PAD inference overhead	15.5 ms (C2→C3 ablation delta)

Table 7 TRIssecure V2 DFA state roles. ★ Accepting state; × trap state (sink).

State	Role and Security Function
IDLE	Awaits NFC card presentation; rate-limit counter initialised.
RATECHECK	Enforces 5-failure/300 s sliding window; blocks brute-force (P6).
NFCVERIFY	PN532 SPI read; UID format validation and database lookup.
VOTERLOOKUP	Decrypts and retrieves voter record; checks structural validity.
ELIGIBILITYCHECK	Atomically reads has_voted ; enforces double-vote prevention (P5).
FACEMATCH	ArcFace cosine similarity against stored encrypted embedding.
FUSION	Computes combined log-LR; applies threshold $\theta=1.0$.
PAD	MiniFASNetV2 liveness classification; blocks presentation attacks.
SESSIONISSUED★	Issues 256-bit token; atomically sets has_voted .
REJECTED×	Sink state; all failure transitions terminate here (P3/P4).

Table 8 Security levels of TRIssecure V2 primitives (bits). † Ed25519/X25519 classical fallback; Kyber/Dilithium form the hybrid layer providing forward secrecy against both adversary classes.

Primitive	Cl.	PQ	Standard
AES-256-GCM (templates)	256	128	FIPS 197
Argon2id (KDF)	256	128	RFC 9106
SHA-256 (receipts)	256	128	FIPS 180
Ed25519 (sign)†	128	×	RFC 8032
X25519 (KEM)†	128	×	RFC 7748
Dilithium-3 (PQC sign)	128	128	FIPS 204
Kyber-768 (PQC KEM)	128	128	FIPS 203

Table 9 Sensitivity of biometric metrics to $\pm 10\%$ Gaussian parameter variation. Nominal: $\sigma_G=0.08$, $\sigma_I=0.12$.

σ_G	σ_I	EER (%)	AUC
0.072 (−10%)	0.108 (−10%)	1.38	0.9993
0.080 (nominal)	0.120 (nominal)	1.55	0.9991
0.088 (+10%)	0.132 (+10%)	1.71	0.9988

Table 10 Biometric performance ($n=1\,000$ genuine/impostor, ArcFace priors).

Metric	Value
EER	1.55% at $\tau = 0.539$
TAR @ 0.1% FAR	89.70%
TAR @ 1.0% FAR	97.70%
ROC AUC	0.9991

Table 11 Threshold sweep (15 operating points). Bold = EER point.

τ	FAR (%)	FRR (%)	TAR (%)
0.100	91.3	0.0	100.0
0.386	16.7	0.0	100.0
0.443	7.8	0.1	99.9
0.500	2.9	0.3	99.7
0.539	1.55	1.55	98.45
0.557	1.0	2.3	97.7
0.614	0.3	9.5	90.5
0.671	0.0	27.3	72.7
0.729	0.0	54.3	45.7

Table 12 Bootstrap 95% percentile CI ($B=10\,000$ replicates).

Metric	Estimate	95% CI
EER (%)	1.55	[1.38, 1.71]
ROC AUC	0.9991	[0.9988, 0.9994]
TAR @ 1% FAR (%)	97.70	[97.41, 97.98]
TAR @ 0.1% FAR (%)	89.70	[89.23, 90.16]

Table 13 PAD metrics per ISO/IEC 30107-3.

Metric	Attack type	Rate (%)
BPCER	—	1.00
APCER (print)	Photo	5.00
APCER (replay)	Video	10.00
APCER (combined)	Both	7.50
ACER	—	4.25

Table 14 Ablation study: cumulative component contribution. [†] at system default threshold $\tau=0.55$.

Config	Components	EER (%)	Eff. FAR (%)	Latency (ms)	Attacks
C1	Face only [†]	1.45	1.2224	95.2	3/12
C2	+Bayesian NFC fusion	1.55	0.0010	95.5	6/12
C3	+PAD liveness	1.55	0.0010	110.7	9/12
C4	+PQC signatures (full)	1.55	0.0010	125.1	12/12

Table 15 Adversarial attack simulation ($n=1000$ per category). Success = attack reaches the voting step. Rates are fraction of n .

Attack	After face threshold	After PAD liveness	After NFC AES-GCM	Final success
Print photo	11.0%	0.7%	0%	0%
Video replay	26.7%	2.2%	0%	0%
Deepfake (est.)	59.7%	14.0%	0%	0%
NFC UID clone	—	—	0%	0%
Session replay	12.0% [†]	—	—	0%
Database tamper	—	—	—	0%
Detection rate (hash chain, database tamper)				100%

[†]Within 60 s TTL; one-time-use flag blocks all replays.

Table 16 Kyber-768/Dilithium-3 benchmarks on Raspberry Pi 4 (Cortex-A72, ARM64). Measured via liboqs 0.15.0 / liboqs-python 0.15.0 ($n=200$ trials, mean $\pm$ 1 SD). Classical fallback (Ed25519/X25519) shown for comparison.

Operation	Standard	Measured (ms)	SD (ms)
ML-KEM-768 keygen	FIPS 203	0.082	0.033
ML-KEM-768 encaps	FIPS 203	0.088	0.015
ML-KEM-768 decaps	FIPS 203	0.095	0.011
ML-DSA-65 keygen	FIPS 204	0.400	0.039
ML-DSA-65 sign	FIPS 204	1.536	1.007
ML-DSA-65 verify	FIPS 204	0.390	0.034
PQC auth path total[†]	—	2.289	—
X25519 ECDH (classical)	RFC 7748	1.022	—
Ed25519 sign+verify (classical)	RFC 8032	0.532	—

[†]Auth path = encaps + decaps + sign + verify; keygen runs at enrolment only. All values measured on-device; no estimated figures.

Table 17 Cryptographic benchmarks on Raspberry Pi 4 (ARM64), 95% CI.

Operation	n	Mean (ms)	95% CI
Argon2id ($t=3$, $m=64$ MB)	20	494.5	[492.7, 496.3]
PBKDF2-HMAC-SHA256 (100k)	100	106.1	[105.1, 107.1]
AES-256-GCM enc. (2 kB)	1000	0.040	[0.040, 0.041]
HMAC-SHA256 (256 B)	10000	0.013	[0.013, 0.013]
Ed25519 sign+verify	200	0.532	[0.527, 0.538]
X25519 keygen+encap+decap	200	1.022	[1.002, 1.042]
SHA-256 chain ($n=1000$)	3	4.27	[4.11, 4.43]
Merkle prove+verify ($n=1000$)	5	4.82	[4.74, 4.90]

Table 18 End-to-end authentication latency on Raspberry Pi 4, $n=50$.

Metric	Value
<i>End-to-end latency</i>	
Mean	110.7 ms
p50	110.7 ms
p95	110.9 ms
p99	111.3 ms
<i>Stage breakdown (mean)</i>	
NFC card read	15.1 ms
Database lookup	0.04 ms
Face embedding (ONNX)	95.2 ms
Cosine match	0.28 ms
Session issuance	0.06 ms
Throughput (sustained)	541.6 voters/min
CPU temperature (load)	52.5 °C

Table 19 Security-layer latency breakdown on Raspberry Pi 4 (ARM64). PAD latency measured via C2→C3 ablation. All other values from Table 16 and Table 17. † Runs post-authentication; not in the interactive critical path.

Security Component	Latency (ms)
<i>Biometric liveness (PAD)</i>	
MiniFASNetV2 ONNX (ablation C2→C3)	15.500
<i>Post-quantum cryptography (FIPS 203/204)</i>	
Kyber-768 encaps + decaps	0.183
Dilithium-3 sign + verify	1.926
PQC auth-path subtotal	2.289
<i>Symmetric encryption & integrity</i>	
AES-256-GCM template decrypt	0.040
HMAC-SHA256 KDF output tag check	0.013
<i>Protocol enforcement</i>	
DFA state-transition check	<0.010
<i>End-to-end verifiability</i>	
Merkle prove + verify ($n=1\,000$) [†]	4.820
In-path security overhead (PAD + crypto + DFA)	17.85
Total overhead (incl. post-auth Merkle)	22.67

Table 20 Proof-of-concept pilot results ($N=20$ participants, real hardware). 95% CI: one-sided Clopper–Pearson upper bound for zero-failure observations; “—” where CI is not applicable.

Metric	Observed	95% CI Upper
<i>Study parameters</i>		
Participants	20	—
Sessions	5 (3 days)	—
Genuine attempts	221	—
Impostor attempts	35	—
Replay attacks	15	—
<i>ISO/IEC 30107-3 metrics</i>		
FAR	0.00%	$\leq 8.20\%$
FRR	0.00%	$\leq 1.35\%$
APCER (print)	n/a	—
APCER (replay)	0.00%	$\leq 18.10\%$
APCER (combined)	0.00%	$\leq 5.82\%$
BPCER	0.00%	$\leq 1.35\%$
ACER	0.00%	$\leq 3.58\%$

Table 21 TRIsure V2 deployment configurations by station size. Assumes 40 s voter interaction time; 6 h peak window.

Station size	Terminals	Voters/hr	Peak capacity
Small (≤ 500 voters)	1	90	540
Medium ($\leq 2\,000$ voters)	3	270	1 620
Large ($\leq 5\,000$ voters)	8	720	4 320